Lecture 18

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Inequality constraints:

$f: \mathbb{R}^n \to \mathbb{R}, h: \mathbb{R}^n \to \mathbb{R}^m, g: \mathbb{R}^n \to \mathbb{R}^r$

$$\min_{x \in \mathbb{R}^n} f(x)$$

s.t. $h(x) = 0$ (

$h_i(x) = 0, i = 1, \cdots, m$)

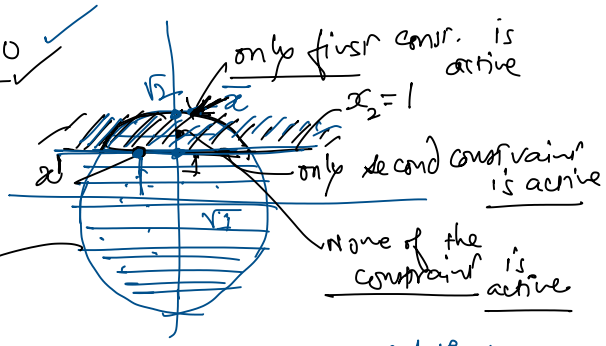
$g(x) \leq 0$ (

$g_j(x) \leq 0, j = 1, \cdots, r$)

Active inequality constraints: For a $x \in \mathbb{R}^n$, the set of active inequality constraints

$A(x) = \{j : g_j(x) = 0\}$

Example $\begin{cases} g_1(x) = x_1^2 + x_2^2 - 2 \leq 0 \\ g_2(x) = 1 - x_2 \leq 0 \\ \Rightarrow x_2 \geq 1 \end{cases}$



$x_1^2 + x_2^2 = 2$

$\bar{x} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix}$

$g_1(\bar{x}) = \bar{x}_1^2 + \bar{x}_2^2 - 2 = 0$ (Active)

$g_2(\bar{x}) = 1 - \bar{x}_2 < 0$ (Not active)

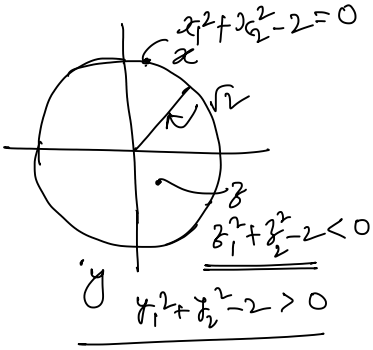
$A(\bar{x}) = \{1\}$

$g_1(\bar{x}) = 0$
but $g_2(x) \neq 0$

$0 > x_1^{1^2} + x_2^{1^2} - 2 = g_1(x^1) < 0$ not active.

$0 = 1 - x_2^1 = g_2(x^1) = 0$ — Active

$A(x^1) = \{2\}$



min $f(x)$

s.t. $x \in S$

x^* is a local minimum : there is a $\epsilon > 0$

s.t.

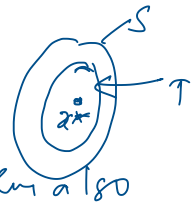
$$f(x)^* \leq f(x) \quad \forall x \in B_\epsilon(x^*) \cap S$$

Another problem

min $f(x)$

s.t. $x \in T$

$T \subseteq S$



Suppose $x^* \in T$

Claim x^* is a local minimum of this problem also

because $f(x^*) \leq f(x) \quad \forall x \in B_\epsilon(x^*) \cap T$.

min $f(x)$

s.t. $h_i(x) = 0 \quad i = 1, \cdots, m$

$g_j(x) \leq 0 \quad j = 1, \cdots, r$

— call it (P1)

$S = \{x \in \mathbb{R}^n : h_i(x) = 0 \quad i = 1, \cdots, m$
$g_j(x) \leq 0 \quad j = 1, \cdots, r\}$

Suppose x^* is a local minimum

$A(x^*)$: set of active constraints at x^*

$$\begin{cases} g_j(x^*) = 0 & \forall j \in A(x^*) \\ g_j(x^*) < 0 & \forall j \notin A(x^*) \end{cases}$$

$$\min f(x) \quad \text{s.t. } h_i(x) = 0 \quad i=1,\dots,m$$
$$g_j(x) \le 0 \quad \forall j \in A(x^*)$$

(P2) $S_1 = \{x \in \mathbb{R}^n : h_i(x) = 0 \quad i=1,\dots,m$
$g_j(x) \le 0 \quad j \in A(x^*)\}$

$S \subseteq S_1$

Claim: x^* is a local minimum of (P2)

Suppose x^* is not a local min. of (P2)

there exists a $\bar{x} \in S_1 \cap B_\epsilon(x^*)$

s.t. $f(\bar{x}) < f(x^*)$

$\bar{x} \in S_1 \iff h_i(\bar{x}) = 0 \quad \forall i$
$g_j(\bar{x}) \le 0 \quad \forall j \in A(x^*)$

$g_j(x) \le 0$ some $j \notin A(x^*)$

$\exists$ a $\bar{x} \in S_1$ s.t. $f(\bar{x}) < f(x^*)$

$\bar{x}$ also satisfies $g_j(\bar{x}) \le 0$

$\Rightarrow x^*$ cannot be a local minimum of (P1)

~~Contradiction~~

So, x^* is a local minimum of (P2)

Consider another problem

(P3) $\begin{cases} \min f(x) \\ \text{s.t. } h_i(x) = 0 \quad i=1,\dots,m \\ g_j(x) = 0 \text{ for } j \in A(x^*) \end{cases}$

$R = \{x \in \mathbb{R}^n : h_i(x) = 0 \quad i=1,\dots,m$
$g_j(x) = 0 \quad \forall j \in A(x^*)\}$

$R \subseteq S_1$

$\Rightarrow x^*$ is a local minimum of (P3)

So it must satisfy: $\exists\ \lambda^* \in \mathbb{R}^m$, μ_j^* $j \in A(x^*)$ s.t.

$$\nabla f(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla h_i(x^*) + \sum_{j \in A(x^*)} \mu_j^* \nabla g_j(x^*) = 0$$

Similarly 2nd order conditions.

Regularity: $\nabla h_1(x^*) \dots \nabla h_m(x^*) \quad \nabla g_j(x^*) \quad \forall j \in A(x^*)$

should be linearly independent.

$m + |A(x^*)|$ vectors

Karush Kuhn Tucker Theorem (necessary conditions for local optimality): Let f and h be continuously differentiable. Let x^* be a local minimum with active inequality constraints [Equation] . Let it be regular with respect to the equality and the active inequality constraints ([Equation] linearly independent). Then there exist unique [Equation] such that [Equation] for all $j \in A(x^*)$,

[Equation]

comes from sensitivity

If in addition [Equation] and g are twice continuously differentiable then $\forall d \in V(x^*)$

[Equation]

for all [Equation] such that [Equation] and [Equation] .

$\forall d: \nabla h_i(x^*)^T d = 0$
$i = 1, \dots, m$
& $\nabla g_j(x^*)^T d = 0$
$\forall j \in A(x^*)$

How do we use this result

inequality constraints: $\{1, 2, \dots, r\}$

Suppose $A \subseteq \{1, 2, \dots, r\}$ is the set of active constraints at local opt.

First order cond $h(x) = 0$ — m equations

$g_j(x) = 0 \quad \forall j \in A$ — $|A|$ equations

$\nabla f(x) + \sum_{i=1}^{m} \lambda_i h_i(x) + \sum_{j \in A} \mu_j g_j(x) = 0$ — n equations

unknowns x, λ, μ
$\# n \quad \# m \quad \# |A|$

Solve the above for (x, λ and μ)

- check 2nd order condition for each feasible (x, λ, μ)
- compare those that satisfy these conditions
- this exercise has to repeated for each $A \subseteq \{1, 2, \dots, r\}$ $\# 2^r$ times
- practical only if r is small.

Approach to obtain x^*: Fix an [Equation] and (a) solve

$h(x) = 0$

[Equation]

$\nabla f(x) + \nabla h(x) \lambda = 0$

for [Equation] unknowns x and λ and [Equation] .

(b) Impose second order necessary conditions if applicable.

(c) Inquire each of the candidate solutions.

Repeat for all [Equation] .

Define $L(x, \lambda, \mu) = f(x) + \sum_{i=1}^{m} \lambda_i h_i(x) + \sum_{j=1}^{r} \mu_j g_j(x)$

$\boxed{\mu_j g_j(x^*) = 0 \quad \forall j \notin A(x^*)}$

$= f(x) + \lambda^T h(x) + \mu^T g(x)$

$g_j(x^*) = 0 \ \forall j \in A(x^*)$
$\Rightarrow \mu_j g_j(x^*) = 0 \ \forall j$

Lagrange function: [Equation]

[Equation]

Necessary conditions: Let x^* be a local minimum with active inequality constraints [Equation]

Let it be regular with respect to [Equation] and [Equation] . Then there exist unique

[Equation] [Equation] such that

[Equation]

[Equation] for all [Equation]

[Equation]

$\mu^* \in R_+^r$
$\mu_j^* \geq 0 \ \forall j = 1, \dots, r$

for all [Equation] such that [Equation] and [Equation] .

- [Equation] [Equation] (referred to as complementary slackness)

Examples (1) $\min \frac{1}{2}(x_1^2+x_2^2+x_3^2)$

s.t. $x_1+x_2+x_3 \le -3$

$g(x) = x_1+x_2+x_3+3$ $\qquad g(x) \le 0$

Lagrangian $L(x,\mu) = \frac{1}{2}(x_1^2+x_2^2+x_3^2) + \mu(x_1+x_2+x_3+3)$

first order condition

$$D_x L(x,\mu) = 0$$

$$\Rightarrow \begin{bmatrix} x_1+\mu \\ x_2+\mu \\ x_3+\mu \end{bmatrix} = 0$$

$$\Rightarrow x_1 = x_2 = x_3 = -\mu$$

Two Cases (a) $A(x) = \phi$

$$x_1+x_2+x_3 < -3 \quad \text{—— (1)}$$

$$\mu = 0$$

$$\Rightarrow x_1 = x_2 = x_3 = 0$$

this contradicts (1) we don't to pursue this

(b) $A(x) = \{1\}$

$$x_1+x_2+x_3 = -3$$

$$\Rightarrow -3\mu = -3$$

$$\Rightarrow \mu = 1$$

$$\Rightarrow x = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} \quad (\text{say } x^*)$$

second order conditions

$$D^2 f(x) + \mu D^2 g(x) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{(Positive definite)}$$

$$d^T(\quad)d \ge 0 \quad \forall d$$

Hence 2nd order necessary condition is also satisfied.

$\Rightarrow$ it is unique candidate for local minimum

Because objective is a convex fn and the constraint set is also convex, x^* is optimal.

Sufficient conditions for local optimality: Let [Equation] and g are twice continuously differentiable and [Equation] and [Equation] be such that

$h(x^*) = 0$

[Equation]

[Equation] (not allowed to be 0 here)

[Equation] for all $j \in A(x^*)$

[Equation] for all [Equation]

[Equation]

for all [Equation] such that [Equation] and [Equation]. Then x^* is a strict local minimum.

first order feasible directions.

Examples ① Previous example

$$x^* = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}, \quad \mu^* = 1 \checkmark$$

$$d^T\left(D^2 f(x^*) + \mu^* D^2 g(x^*)\right) d > 0 \quad \forall\, d$$

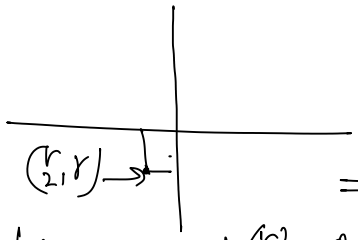
Hence sufficient conditions are satisfied.

$\Rightarrow x^*$ is a strict optimum

② $\min \frac{1}{2}(x_1^2 - x_2^2)$

s.t. $x_2 \le 0$

$g(x) = x_2$



$r < 0$

$$f\left(\tfrac{r}{2}, r\right) = \frac{1}{2}\left(\frac{r^2}{4} - r^2\right) = -\frac{3r^2}{8}$$

$\inf_{x_2 \le 0} f(x) = -\infty$, no opt. soln exists.

Lagrangian

$$L(x, \mu) = \frac{1}{2}(x_1^2 - x_2^2) + \mu x_2$$

First order condition

$$D_x L(x, \mu) = \begin{bmatrix} x_1 \\ -x_2 + \mu \end{bmatrix}$$

Setting to 0

$$x = \begin{bmatrix} 0 \\ \mu \end{bmatrix}$$

Two cases ① $A(x) = \phi$

$\mu = 0$

$x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$g\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) = 0$ Contradicts that $A\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) = \phi$

② $A(x) = \{1\}$

$g\left(\begin{bmatrix} 0 \\ \mu \end{bmatrix}\right) = 0$

$\Rightarrow \mu = 0$

only $x^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\mu^* = (0)$ satisfy first order necessary conditions.

2nd order condition

$$D^2_{xx} L(x, \mu) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Need to show: $d^T D^2_{xx} L(x, \mu) d > 0 \quad \forall\, d \ne 0$ and first order feasible dirn

First order feasible directions

$$\nabla g(x)^T d = 0$$

$$\Rightarrow \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = 0$$

$\rightarrow d_2 = 0$

$\rightarrow \alpha_2 \sim 0$

Hence $\begin{bmatrix} d_1 \\ 0 \end{bmatrix}$ are first order feasible

$$[d_1 \;\; 0] \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} d_1 \\ 0 \end{bmatrix} = [d_1 \;\; 0] \begin{bmatrix} d_1 \\ 0 \end{bmatrix} = d_1^2 > 0 \quad \forall d \neq 0$$

Hence second order sufficient conditions are also satisfied

but $\mu_j^* > 0 \;\; \forall j \in A(x^*)$ is ~~not satisfied~~!

$$\left.\begin{aligned} \min_{x \in \mathbb{R}^n} \; & f(x) \\ \text{s.t.} \; & h(x) = 0 \\ & g_1(x) \leq 0 \\ & g_2(x) < 0 \end{aligned}\right\} =$$

define $S = \{x \in \mathbb{R}^n : g_2(x) < 0\}$

$$\left.\begin{aligned} \min_{x \in S} \; & f(x) \\ & h(x) = 0 \\ & g_1(x) < 0 \end{aligned}\right\} =$$

so far $S = \mathbb{R}^n$

- we now focus on such problems.

Global optimality: [Equation] .

$$\begin{aligned} \min_{x \in S} \; & f(x) \\ \text{s.t.} \; & h(x) = 0 \; (h_i(x) = 0, i = 1, \cdots, m) \\ & g(x) \leq 0 \; (g_j(x) \leq 0, j = 1, \cdots, r) \end{aligned}$$

Lagrange function: [Equation]

[Equation]

Sufficient conditions: If [Equation] satisfy

[Equation] (global minimization)

$h(x^*) = 0$

[Equation]

[Equation] [Equation]

then x^* is a global minimum.